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(9630) PH04 Energy and Energy Resources Report on the examination

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REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH04 ENERGY AND ENERGY RESOURCES – JANUARY 2023

This paper had a similar level of challenge based on those topics and skills examined in previous papers. As seen in many past series, calculations were generally done very well whereas written answers often lacked detail (eg **03.1, 05.2**) or precision (eg **04.1, 05.8**). There were notable gaps in knowledge and understanding about aspects of fission and fusion.

Students were successful throughout Section B.

SECTION A

QUESTION 01

01.1 About three-quarters of the cohort could read and interpret the log scales correctly.

01.2 Many students were familiar with a method to test whether a data set support a particular relationship, in this case the inverse-square law. The most common approach was to show that the product Id^2 was almost constant for the three pairs of data. A commonly seen error was for students to interpret the symbol I for electric current and to use either $P = VI$ or $P = I^2R$ to find some pattern in the data.

01.3 Just over one fifth of the cohort gave a relevant procedure in this question, but almost none gave a supporting explanation. The most common answer related to keeping the potential difference across, or current in, the lamp the same.

QUESTION 02

02.1 Students were expected, for the first marking point, to show how some aspect of **Figure 4** would lead to an angular displacement of approximately $2.6\ \text{rad}$. Most students achieved this by marking and measuring an angle of approximately $150°$.

02.2 Nearly 75% of the cohort gained this mark.

02.3 This question was correctly answered by about 80%. A common error, for only one mark maximum, was to use an angle of $\pi\ \text{rad}$ for the first rotation. A frequently seen response, for no credit, was to multiply $3.6\ \text{rad s}^{-2}$ by $1.2\ \text{s}$.

02.4 This question differentiated well. Many students supported their arguments algebraically, which was acceptable. Many students misinterpreted how the moments of inertia would be different for the wheel and the disc.

QUESTION 03

03.1 Most students struggled to give a suitable practical problem involved in sustaining fusion within the reactor. At A-level standard, students are expected to give more developed responses than, for instance, '*high temperatures*'.

03.2 About 80% of the cohort coped well with this calculation and achieved full marks. The large number of significant figures in the data proved difficult for some students but they were given credit for incorrect calculations when their working out was evident.

03.3 Students are expected, for this specification, to know the hydrogen cycle. As a 'deduce' question, students were expected to give the reasons for the number of times for each of the reactions, rather than simply state the result.

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QUESTION 04

04.1 There was a good distribution of marks for this question. Incorrect responses often referred to an ability (when energy demand is below supply availability) to store extra ''*electricity*'' that could be used later.

04.2 This calculation was done accurately by about 60% of the cohort. Unusually, some students used 1.3×10^3 for the density of air, which was not condoned.

04.3 Few students had a problem with this energy-transfer calculation. Those who applied a '*suvat*' (kinematic equation) approach did not get credit. Students should be reminded that '*suvat*' equations are valid only when the acceleration is constant.

04.4 This question discriminated well. Students achieved credit most commonly by addressing the last two marking points: referring to a force on the blade due to the water and linking this to the production of a torque. An isolated statement of '$T=Fr$' did not get credit. The origin of the force on the blade, in terms of the momentum change, was less frequently mentioned.

QUESTION 05

05.1 There was a reasonable distribution of marks here. As with other occurrences of the 'deduce' command, a reason was required for the final marking point. There were several numerical routes available, and students needed to state explicitly how their chosen method demonstrated that the decay was not possible.

05.2 About 30% of the cohort did not describe a chain reaction in even a basic manner and only about 40% could give a simple description of a chain reaction. Few gave specific details of the process. For example, they failed to reference a nucleus, preferring terms such as '*uranium fuel*' – a high level of appropriate technical language is expected at this standard.

05.3 Just over 70% of the cohort gave an appropriate substance for a moderator.

05.4 Students performed very poorly in this question, with just short of 10% gaining a mark. Very few demonstrated an understanding that the average kinetic energy of the neutrons was comparable to that of the atoms of the moderator and fuel. Common incorrect responses for the term were that the neutrons caused heating or that they had a very high kinetic energy.

05.5 Students did slightly better on this question than on **05.4**. Most students gained a mark for relating the lower neutron speed to the increased probability of a further fission. However, few could link this to the condition needed for the chain reaction. A large majority of the cohort had the notion that the high speed of the neutron meant a high reaction rate and so slowing the neutron speed was necessary to reduce the overall power output from the reactor.

05.6 Most students handled this calculation well. Those who were unsure how to deal with the exponent were wise to show their working and thus gained partial credit.

05.7 About 60% of students handled this multi-stage internal energy calculation without difficulty. A curious common mistake was the conversion of the latent heat of fusion from $\mathrm{kJ\ kg^{-1}}$ to $\mathrm{J\ kg^{-1}}$ but with no equivalent conversion for the specific heat capacity values.

05.8 Very few students could give any problem suitable for A-level standard in this question. Statements such as '*store for a long time*' or '*keep away from people*' were not detailed enough. Heating due to the beta emission would require cooling of the waste; there would be little heating due to gamma emissions. The need for a container of thick (lead/steel/concrete) walls had to be linked to the gamma radiation; such a container would not be necessary for the beta radiation.

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QUESTION 06

06.1 A significant number of students failed to any gain marks in this question. Many did not reference an ideal gas equation, as directed in the command, at the beginning of their explanation. Students who made useful progress often did so using terms with subscript notation (p_1, T_1, V_1) in the ideal gas equation. For the second marking point, students often referred to $\boldsymbol{p = \frac{m}{V}}$ to explain why the increase in density would produce a decrease in volume and so $\boldsymbol{\frac{V_1}{V_2} < 1}$.

06.2 This question discriminated well, with an even spread of marks. As with **06.2**, many students failed to follow the command and reference the first law of thermodynamics in their answer. The important details to address were: there was a compression; and it was rapid. The first point meant that work was done on the gas (by the compressor blades). The second point meant there was very little time for the gas to cool by heat transfer.

06.3 This question was well answered with over 80% of the cohort achieving full marks.

06.4 Nearly 40% of students gained full marks in this question. Two simultaneous equations involving the rate of energy transfer by conduction were required. Some students appreciated the need to set up simultaneous equations but did so in terms of $\boldsymbol{UA\Delta\theta}$.

06.5 Over 60% of the cohort achieved full marks in this question. A common error was trying to calculate the torque using $\boldsymbol{T = I\alpha}$. It may be that some students think that any reference to a torque is always to a resultant torque. A comparison with the analogous equation $\boldsymbol{P = Fv}$ may be useful in this regard.

06.6 Just over half of the cohort failed to appreciate that the total area under the curve was equal to the work done by the gas. Many attempted to use $\boldsymbol{W = p\Delta V}$, which is only applicable at a constant pressure. An answer of 1×10^8 to one significant figure (from correct working) was condoned, but a bald '10^8' was not.

SECTION B

Overall, students did not find any questions in Section B too challenging.

Questions 7, 9, 11, 12, 13, 15, 17, 18 and 19 were particularly well done.

In two questions, 14 and 20, the most popular response was not the answer.

In question 14, option B was the most popular response whereas the answer was A. Option B includes the possibility of an isothermal compression, which would not increase the internal energy of a fixed mass of gas.

In question 20, option B was the most popular response, whereas the answer was A. Option B is obtained by taking moments about the midpoint of the rod, not about the centre of mass.

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